$\boxed{P\,4.1}$ (a) denote $r = (x, y, z)$ so that $r_1 = x, r_2 = y, r_3 = z$

Thus $[r_i, r_j] = r_i r_j - r_j r_i = 0$

Then $[p_x, p_y] f = \left(\frac{\hbar}{i}\frac{\partial}{\partial x}\left(\frac{\hbar}{i}\frac{\partial}{\partial y}\right) - \frac{\hbar}{i}\frac{\partial}{\partial y}\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)\right) f =$

$$= \left(\frac{\hbar}{i}\right)\left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x}\right) = 0 \Rightarrow [p_i, p_j] = 0$$

As in 1D $[x, p_x] f = \left(x \frac{\hbar}{i}\frac{\partial}{\partial x} - \frac{\hbar}{i}\frac{\partial}{\partial x} x\right) f =$

$$= \frac{\hbar}{i}\left(x \frac{\partial f}{\partial x} - \frac{\partial}{\partial x}(xf)\right) = -\frac{\hbar}{i} f = i\hbar f$$

$$\Rightarrow [x, p_x] = i\hbar$$

Similarly $[y, p_y] = [z, p_z] = i\hbar$.

Finally, $[y, p_x] f = \left(y \frac{\hbar}{i}\frac{\partial}{\partial x} - \frac{\hbar}{i}\frac{\partial}{\partial x} y\right) f =$

$$= \frac{\hbar}{i}\left(y \frac{\partial f}{\partial x} - \frac{\partial}{\partial x}(yf)\right) = \frac{\hbar}{i}\left(y \frac{\partial f}{\partial x} - y\frac{\partial f}{\partial x}\right) = 0$$

$$\Rightarrow [r_i, p_j] = i\hbar\, \delta_{ij}.$$

(b) $\frac{d}{dt}\langle \vec{r} \rangle = \frac{1}{m}\langle \vec{p} \rangle$

because $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

Consider, e.g., $\frac{d}{dt}\langle x \rangle = \frac{i}{\hbar}\langle [H, x] \rangle$

$$H = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(r) \Rightarrow$$

$$[H, x] = \frac{1}{2m}[p_x^2, x] = \frac{1}{2m}\left(p_x [p_x, x] + [p_x, x] p_x\right) =$$

$$= -\frac{i\hbar}{m} p_x$$

and all other terms commute because of (a).

$$\Rightarrow \frac{d}{dt}\langle \vec{r} \rangle = \frac{i}{\hbar}\left(-\frac{i\hbar}{m}\right)\langle p_x \vec{i} + p_y \vec{j} + p_z \vec{k} \rangle = \frac{1}{m}\langle \vec{p} \rangle.$$

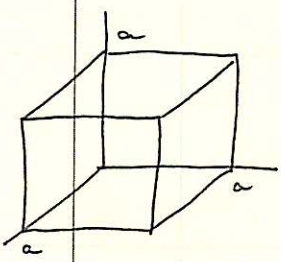
Similarly, $\frac{d}{dt}\langle p_x\rangle = \frac{i}{\hbar}\langle [H, p_x]\rangle$

But $[H, p_x] = [V(r), p_x] = i\hbar \frac{\partial V}{\partial x}$ (see eq. 3.65)

Thus $$\frac{d}{dt}\langle \vec{p}\rangle = \frac{i}{\hbar}(i\hbar)\left\langle \frac{\partial V}{\partial x}\vec{i} + \frac{\partial V}{\partial y}\vec{j} + \frac{\partial V}{\partial z}\vec{k}\right\rangle$$

$$= \left\langle -\vec{\nabla} V(r)\right\rangle .$$

$\boxed{\text{P4.2}}$ Particle in a 3D box.



$$V = \begin{cases} 0, & x, y, z \in \text{box of side } a. \\ \infty & \text{otherwise} \end{cases}$$

Inside the box: $-\frac{\hbar^2}{2m}\left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2}\right)\psi = E\psi$

Outside: $\psi = 0$ because $V = \infty$.

Separation of variables: $\psi = XYZ = X(x)\,Y(y)\,Z(z)$

$$\Rightarrow YZ\frac{d^2X}{dx^2} + XZ\frac{d^2Y}{dy^2} + XY\frac{d^2Z}{dz^2} = -\frac{2mE}{\hbar^2}XYZ$$

Divide by XYZ:

3 one-dimensional boxes!

$$\underbrace{\frac{1}{X}\frac{d^2X}{dx^2}}_{-k_x^2} + \underbrace{\frac{1}{Y}\frac{d^2Y}{dy^2}}_{-k_y^2} + \underbrace{\frac{1}{Z}\frac{d^2Z}{dz^2}}_{-k_z^2} = \underbrace{-\frac{2mE}{\hbar^2}}_{-k^2}$$

$$\Rightarrow E = \frac{\hbar^2}{2m}k^2 = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right)$$

Consider, e.g., x-axis: $\frac{d^2X}{dx^2} = -k_x^2 X$

$$\Rightarrow X(x) = A\cos k_x x + B\sin k_x x$$

Boundary condition: $X(0) = X(a) = 0 \Rightarrow A = 0,\ \sin(k_x a) = 0$

$\Rightarrow \quad k_x a = \pi n_x \quad , \; k_x = \frac{\pi n_x}{a}$

Normalization (done before, see Ch. 2.2) $B = \sqrt{\frac{2}{a}}$

Thus $\psi(x,y,z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{\pi n_x}{a} x\right) \sin\left(\frac{\pi n_y}{a} y\right) \sin\left(\frac{\pi n_z}{a} z\right)$.

$$E = \frac{\pi^2 \hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

(b) The lowest energy state $n_x = n_y = n_z = 1$:

particle is in the ground state in each 1D box!

$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}(1+1+1) = 3E_0$, where $\boxed{E_0 = \frac{\hbar^2 \pi^2}{2ma^2}}$

This state is unique: $d = 1$.

Excited states are obtained by increasing one of n's by 1

n_x	n_y	n_z	$n_x^2+n_y^2+n_z^2$	Degeneracy	Energy	
1	1	1	3	1	$3E_0$	E_1
1	1	2	6	3	$6E_0$	E_2
1	2	1				
2	1	1				
1	2	2	9	3	$9E_0$	E_3
2	1	2				
2	2	1				
1	1	3	11	3	$11E_0$	E_4
1	3	1				
3	1	1				
2	2	2	12	1	$12E_0$	E_5
1	2	3	14	6	$14E_0$	E_6
1	3	2				
3	2	1				
2	3	1				
2	1	3				
2	3	1				

(c) Continue with the Table, but (to save time only) we can skip degeneracy question until we get to 14TH level, E_{14}.

n_x	n_y	n_z	$n_x^2+n_y^2+n_z^2$	Energy
1	1	1	3	E_1
1	1	2	6	E_2
1	2	2	9	E_3
1	1	3	11	E_4
2	2	2	12	E_5
1	2	3	14	E_6
2	2	3	17	E_7
1	1	4	18	E_8
1	3	3	19	E_9
1	2	4	21	E_{10}
2	3	3	22	E_{11}
2	2	4	24	E_{12}
1	3	4	26	E_{13}
3	3	3	27	E_{14} !
1	1	5	27	

The state (333) is unique, but (115) is 3-fold degenerate:

$$\left.\begin{matrix}115\\151\\511\end{matrix}\right\} \Rightarrow \text{degeneracy of } E_{14} \text{ is } 4$$

$$3^2+3^2+3^2 = 1^2+1^2+5^2$$

this is example of <u>accidental</u> degeneracy.

$\boxed{P4.3}$ $\quad Y = \Theta \phi$

$$\phi(\varphi) = e^{im\varphi}, \quad \Theta(\theta) \sim P_\ell^m(\cos\theta)$$

$\ell = m = 0$

$$Y_0^0(\theta,\varphi) = \sqrt{\frac{1}{4\pi}}\, P_0^0(\cos\theta)$$

$$P_0^0(x) = P_0(x), \quad P_0(x) = 1 \quad \Rightarrow \quad Y_0^0 = \frac{1}{\sqrt{4\pi}}.$$

Normalization: $\displaystyle\int_0^\pi \frac{1}{4\pi} \sin\theta\, d\theta \int_0^{2\pi} d\varphi = 1.$

$\ell = 2$
$m = 1$

$$Y_2^1 = (-1)^1 \sqrt{\frac{2\cdot 2+1}{4\pi}\,\frac{(2-1)!}{(2+1)!}}\; e^{i\varphi} P_2^1(\cos\theta)$$

$$P_2^1(x) = (1-x^2)^{1/2}\left(\frac{d}{dx}\right) P_2(x)$$

$$P_2(x) = \frac{1}{2^2\cdot 2!}\left(\frac{d}{dx}\right)^2 (x^2-1)^2 = \frac{1}{8}\frac{d^2}{dx^2}(x^4-2x^2+1) =$$

$$= \frac{1}{8}\left(4\cdot 3\, x^2 - 2\cdot 2\right) = \frac{1}{2}(3x^2-1)$$

$$\Rightarrow P_2^1(x) = (1-x^2)^{1/2}\frac{d}{dx}\,\frac{1}{2}(3x^2-1) = 3x(1-x^2)^{1/2}$$

$$\Rightarrow P_2^1(\cos\theta) = 3\cos\theta\sqrt{1-\cos^2\theta} = 3\cos\theta\sin\theta$$

$$\Rightarrow Y_2^1(\theta,\varphi) = (-1)\sqrt{\frac{5}{4\pi}\cdot\frac{1}{6}}\; 3e^{i\varphi}\cos\theta\sin\theta$$

$$= -\sqrt{\frac{15}{8\pi}}\, e^{i\varphi}\sin\theta\cos\theta$$

Normalization:

$$\int |Y_2^1|^2 \sin\theta\, d\theta\, d\varphi = 2\pi\left(\frac{15}{8\pi}\right)\int_0^\pi \sin^2\theta\cos^2\theta\,\sin\theta\, d\theta$$

$$= \frac{15}{4}\int_1^{-1}(-)(1-\cos^2\theta)\cos^2\theta\, d(\cos\theta) = \frac{15}{4}\int_{-1}^{1}(1-x^2)x^2\,dx =$$

$-\sin\theta\, d\theta = d(\cos\theta)$ $\qquad X = \cos\theta$

$$\int \left(Y_2^1\right)^2 \sin\theta\, d\theta\, d\varphi = \frac{15}{4}\cdot 2\cdot \int_0^1 \left(x^2 - x^4\right) dx =$$

$$= \frac{15}{2}\left(\frac{1}{3}x^3 - \frac{1}{5}x^5\right)\Big|_0^1 = \frac{15}{2}\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{15}{2}\,\frac{5-3}{15} = 1.$$

<u>Orthogonality</u> $\int \left(Y_0^0\right)^* Y_2^1 \sin\theta\, d\theta\, d\varphi =$

$$= \frac{1}{\sqrt{4\pi}}\left(-\sqrt{\frac{15}{8\pi}}\right)\int_0^\pi \sin\theta\cos\theta\sin\theta\, d\theta \underbrace{\int_0^{2\pi} e^{i\varphi}\, d\varphi}_{(-i)\,e^{i\varphi}\big|_0^{2\pi} = 0} = 0$$

$\boxed{P4.7}$ Eq. 4.46: $n_\ell(x) = -(-x)^\ell \left(\frac{1}{x}\frac{d}{dx}\right)^\ell \frac{\cos x}{x}$

$$n_1(x) = -(-x)\,\frac{1}{x}\frac{d}{dx}\,\frac{\cos x}{x} = \frac{d}{dx}\,\frac{\cos x}{x} =$$

$$= -\frac{\sin x}{x} - \frac{\cos x}{x^2}$$

$$n_2(x) = -x^2\left(\frac{1}{x}\frac{d}{dx}\right)\left(\frac{1}{x}\frac{d}{dx}\right)\frac{\cos x}{x} =$$

$$= -x\,\frac{d}{dx}\,\frac{1}{x}\left(-\frac{\sin x}{x} - \frac{\cos x}{x^2}\right) = x\,\frac{d}{dx}\left(\frac{\cos x}{x^3} + \frac{\sin x}{x^2}\right) =$$

$$= x\left(-\frac{\sin x}{x^3} - \frac{3\cos x}{x^4} + \frac{\cos x}{x^2} - \frac{2\sin x}{x^3}\right) =$$

$$= -\left(\frac{3\sin x}{x^2} + \frac{3\cos x}{x^3} - \frac{\cos x}{x}\right).$$

(b) $n_1(x) \underset{x\to 0}{\to} \frac{-x}{x} - \frac{1}{x^2} = -\frac{1}{x^2} \underset{x\to 0}{\to} \infty$

$$n_2(x) \underset{x\to 0}{\to} (-)\left(\frac{3x}{x^2} + \frac{3}{x^3} - \frac{1}{x}\right) = \frac{-3}{x^3} \underset{x\to 0}{\to} \infty$$

P4.38 (a) $V(r) = \frac{1}{2} m\omega^2 r^2 = \frac{1}{2} m\omega^2 (x^2+y^2+z^2)$

$$\Rightarrow \text{S.E.} \quad -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi + \frac{1}{2} m\omega^2(x^2+y^2+z^2)\psi = E\psi \quad (1)$$

Separation of variables: $\psi(x,y,z) = X(x)\,Y(y)\,Z(z)$

Substitute in (1) and divide then by XYZ:

$$\left(-\frac{\hbar^2}{2m}\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{2}m\omega^2 x^2\right) + \left(-\frac{\hbar^2}{2m}\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + \frac{1}{2}m\omega^2 y^2\right)$$

$$+\left(-\frac{\hbar^2}{2m}\frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} + \frac{1}{2}m\omega^2 z^2\right) = E$$

Clearly, every bracket must be a constant $\Rightarrow$

$$-\frac{\hbar^2}{2m}\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{2}m\omega^2 x^2 = E_x$$

Multiply by X:

$$-\frac{\hbar^2}{2m}\frac{d^2X}{dx^2} + \frac{1}{2}m\omega^2 x^2 X = E_x X$$

Similarly:

$$-\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2} + \frac{1}{2}m\omega^2 y^2 Y = E_y Y$$

$$-\frac{\hbar^2}{2m}\frac{d^2Z}{dz^2} + \frac{1}{2}m\omega^2 z^2 Z = E_z Z$$

three one-dimensional oscillators!

and $E_x + E_y + E_z = E$.

Energies of 1D oscillators are well known: (eq. 2.61)

$$E_x = \hbar\omega\left(n_x+\tfrac{1}{2}\right) \quad n_x = 0,1,2\ldots$$
$$E_y = \hbar\omega\left(n_y+\tfrac{1}{2}\right) \quad n_y = 0,1,2\ldots$$
$$E_z = \hbar\omega\left(n_z+\tfrac{1}{2}\right) \quad n_z = 0,1,2\ldots$$

$$\Rightarrow E_n = \hbar\omega\left(n_x+n_y+n_z+\tfrac{3}{2}\right) = \hbar\omega\left(n+\tfrac{3}{2}\right)$$

where $n = n_x + n_y + n_z = 0,1,2,3\ldots$ integer.

(b) Degeneracy of n-th energy level $E_n = \hbar\omega\left(n+\frac{3}{2}\right)$.

That is, how many ways (partitions) there are to represent integer $\underline{n}$ as a sum of three integers: n_x, n_y, n_z?

$n = n_x + n_y + n_z$ is fixed.

# of Partitions:

1 $\underline{n_x = n}$: $n_y = n_z = 0$

2 $\underline{n_x = n-1}$: $n_y = 1$, $n_z = 0$ <u>OR</u> $n_y = 0$, $n_z = 1$

3 $\underline{n_x = n-2}$: $n_y = 2$ $n_z = 0$ <u>OR</u> $n_y = 0$, $n_z = 2$
<u>OR</u> $n_y = 1$, $n_z = 1$

4 $\underline{n_x = n-3}$: $n_y = 3$ $n_z = 0$ <u>OR</u> $n_y = 0$ $n_z = 3$
<u>OR</u> $n_y = 2$ $n_z = 1$ <u>OR</u> $n_y = 1$ $n_z = 2$

$\vdots$

Clearly,

$$d(n) = \underbrace{1 + 2 + 3 + 4 \ldots + (n+1)}_{(n+1)\text{ terms}} = \frac{1}{2}(n+1)(1+n+1)$$

$$= \frac{1}{2}(n+1)(n+2)$$